

ANUP KUMAR JENA

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SUMMARY

Computer Science undergraduate specializing in Cybersecurity and IoT with hands-on experience building **full-stack web applications using React, Node.js/Express, and FastAPI**. Delivered production-deployed systems with **real-time WebSocket pipelines, REST API design, and Dockerized CI/CD deployments**. Familiar with **microservices architecture, SQL/NoSQL databases, and scalable backend engineering**. Actively learning **Go (Golang)** to expand backend capability. Eager to contribute to **high-throughput, low-latency product development** in a fast-paced startup environment.

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js (learning), Vite, Tailwind CSS, JavaScript, TypeScript, WebSockets
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST API Design, Microservices Architecture, Go (Golang, in progress)
- **Databases:** MySQL, MongoDB, MongoDB Atlas, PostgreSQL, SQL Query Optimization, Redis (basic caching)
- **DevOps & Cloud:** Docker, GitHub Actions, CI/CD Pipelines, Containerization, Vercel, Microsoft Azure (familiar), AWS/GCP (familiar)
- **Testing & Quality:** Jest (basic), API testing with Postman, structured logging, error handling
- **Languages:** JavaScript, TypeScript, Python, Java, C, Bash/Shell Scripting
- **Other:** Git, Agile methodology, Linux/Unix, IoT Systems, LLM API integration

PROJECTS

EcoFlow — Smart Waste Management Platform

[GitHub](#) | [Live Demo](#)

Tech: React (Vite), Node.js, Express.js, MySQL, WebSockets, Docker, Leaflet, OpenAI API, GitHub Actions

- Built a **full-stack waste management system** with a React/Vite frontend and a Node.js/Express backend, deployed live on Vercel with CI/CD via GitHub Actions.
- Engineered **real-time IoT telemetry streaming** using WebSockets, enabling live vehicle and bin status updates on an interactive Leaflet map dashboard.
- Designed a **normalized MySQL schema** supporting vehicle fleet management, bin tracking, driver auth, and KPI analytics endpoints consumed by the React frontend.
- Integrated **OpenAI-powered assistant** for operational query resolution and implemented **JWT-based authentication** with bcrypt password hashing.
- Containerized full application stack with **Docker**, maintaining consistent dev/prod environments aligned with production deployment best practices.

HydroSense — IoT Water Monitoring Platform

[GitHub](#)

Tech: Node.js, Express.js, React, MongoDB Atlas, ESP32, Telegram Bot API, Razorpay, WebSockets

- Architected an **end-to-end IoT data pipeline**: ESP32 sensor telemetry over HTTP, real-time transformation and validation, persistence to MongoDB Atlas, and live React dashboard.
- Built **scalable RESTful API layer** with Node.js/Express for sensor data ingestion, user management, and subscription-gated access via Razorpay payment integration.
- Implemented **threshold-based alerting system** using Telegram Bot API, enabling proactive anomaly notification with sub-second response from data ingestion to alert dispatch.

BankShield — Fraud Detection API Platform

[GitHub](#)

Tech: Python, FastAPI, Docker, REST APIs, Anomaly Detection

- Developed a **production-grade FastAPI backend** for real-time transaction fraud detection with tiered difficulty modes, structured logging, and Dockerized deployment.
- Designed **RESTful API endpoints** for alert retrieval, fraud classification, and result reporting with input validation, error handling, and consistent response contracts.
- Deployed on **Hugging Face Spaces** with Docker; implemented automated testing workflows integrated into the CI pipeline.

MALO — Multi-Agent LLM Orchestration Platform

[GitHub](#) | [Live Demo](#)

Tech: Python, FastAPI, Streamlit, Ollama (local LLM), Multi-Agent Systems, REST APIs

- Built a **FastAPI backend orchestrating 6 AI agents** in parallel with stateful task management, surfacing processing errors and agent health via RESTful endpoints.
- Integrated **local LLM inference** (Ollama) with controlled latency and implemented structured error handling for production-stable, SLA-adherent response pipelines.

EDUCATION

B.Tech in Computer Science and Engineering (IoT & Cybersecurity)

August 2023 – May 2027

Cambridge Institute of Technology, Bengaluru, India

- Specialization in IoT Systems, Cybersecurity, Linux, Computer Networking, and Distributed Systems.

ACHIEVEMENTS & CERTIFICATIONS

- **OpenEnv India 2026** — Ranked Top 800 among 31,000+ participating teams nationwide.
- **2nd Prize Winner** — INNOVEX 2.0 technical innovation competition.
- **Copado Certified AI Professional** — DevOps and AI automation certification (*Jun 2025*).
- **IBM AI Literacy Certified** — Foundational AI concepts and responsible AI practices.